

VANESSA ROBLEDÓ

- Msc Environmental Engineer • PhD Candidate UIOWA •
- Research Assistant at IHHR- Hydroscience & Engineering •
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 - [Research Gate](#) [ORCID](#) [Google Scholar](#)

RESEARCH INTEREST

I am an environmental engineer with a Master's degree in Environmental Engineering from the Universidad de Antioquia, Colombia, and currently pursuing a Ph.D. at the University of Iowa, United States. My research aims to enhance quantitative precipitation forecasting to improve flash flood predictions by integrating insights from storm behavior, specifically mesoscale convective systems (MCSs), and atmospheric processes into nowcasting techniques. I am particularly interested in how the atmospheric environment, from the mesoscale to the synoptic scale, influences the predictability of MCSs and how these systems may evolve under climate change.

EDUCATION

- **Aug 2023 - Present: PhD in Civil and Environmental Engineering.**
Thesis: *Predictability of precipitation forcings for flash flood forecasting.*
University of Iowa, United States.
- **Aug 2020 - Mar 2023: Msc in Environmental Engineering with emphasis in Atmospheric Sciences.**
Thesis: *Spatio-temporal characteristics of mesoscale convective systems over northwestern South America using a new persistence tracking technique.*
University of Antioquia, Colombia.
- **2012 - 2019: Bachelor in Environmental Engineering.**
Academic project: *Validation of GOES-East satellite solar irradiance data over Medellín, Colombia.*
University of Antioquia, Colombia.

RESEARCH EXPERIENCE

INDIVIDUAL-BASED EXPERIENCE

- **2025-Present: Future investigator in NASA Earth and Space Sciences and Technology**
Leader of the project "Predictability of Precipitation Forcings for Flash Flood Forecasting" grant #80NSSC25K0649.

TEAM-BASED EXPERIENCE

- **2023-Present: Research Assistant, Iowa Flood Center.**
 1. **NASA's SERVIR program-West Africa:** Responsible for building and calibrating hydrologic models, developing an operational flash flood forecasting system for West Africa, and conducting capacity-building activities for stakeholders and decision-makers.
 2. **World Meteorological Organization - Flash Flood forecasting system for the Caribbean Region:** Responsible for developing an operational flash flood forecasting system, and conducting capacity-building activities for stakeholders and decision-makers.
- **2022-2023: Research scientist, SIATA: Early Warning System of the Aburra Valley, Colombia.**
Support in the analysis of hydrometeorological data for disaster risk management in Medellín and Aburrá Valley, Colombia.
- **2020-2023 Research Assistant, Univeristy of Antioquia.**
Implementation of a short-term weather research and forecasting system using the WRF model to support command and control systems of the Colombian Air Force, developing a storm tracking algorithm and capacity building on its use.

HONORS & AWARDS

- **Selected Participant, ASP Summer Colloquium, NSF NCAR, Boulder, CO (July 2026).**
- **Selected Participant, CIROH HydroLearn Workshop and Hackathon, (May 2026).**
- **NASA ROSES-24 FINESST Grant (2025).**
- **Iowa Flood Center Excellence in Research Award (2025).**
- **Student Employee of the Year Nominee, University of Iowa (2025).**
- **Bob and Joan Wubbena UI Graduate Engineering Fellowship, University of Iowa (2023–2024).**
- **Eos Editor's Highlight, AGU (2025):** "Mesoscale Convective Systems tracking Method Intercomparison (MCSMIP): Application to DYAMOND Global km-scale Simulations". URL: <https://eos.org/editor-highlights/bringing-storms-into-focus>
- **WMO/GEWEX Regional Hydroclimate Project ANDEX Research Highlight (2024):** "Climatological Tracking and Lifecycle Characteristics of Mesoscale Convective Systems in Northwestern South America". URL: <https://www.andex-rhp.org/?p=4279>

PROFESSIONAL SERVICE

- **Presenter, Hydrologic–Weather Briefing**

Prepare and deliver weekly briefings on hydrologic and meteorological conditions for Iowa Flood Center researchers and staff.

- **Manuscript Reviewer**

Journal of Geophysical Research: Atmospheres, Geophysical Research Letters, Earth and space science.

WORKSHOPS DEVELOPED AND DELIVERED

- **2025 - NASA Marshall Space Flight Center, Huntsville, Alabama:** Capacity-building workshop on the EF5 model for operational flash flood forecasting (Presenter).
- **2025 - AGRHYMET Regional Centre (ARC), Accra, Ghana:** Capacity-building workshop on the EF5 model for operational flash flood forecasting (Presenter).
- **2026 - World Meteorological Organization Workshop for the Caribbean region:** Capacity-building Workshop on Hydrological Modeling of Flash Floods with EF5 (Presenter).

SELECTED CONFERENCE CONTRIBUTIONS

- **Robledo, V.,** Vergara, H., & Abdelkader, M. (2025). Assessing the Role of Satellite-Based Mesoscale Convective System Climatology for Enhancing Flash Flood Forecasts. In 106th AMS Annual Meeting. Houston, USA.
- **Robledo, V.,** Vergara, H., Abdelkader, M., Kruczkiewicz, A. (2025). Towards a Global High-Resolution Flash Flood Climatology: Leveraging Simulation and Multi-Source Validation. In AGU Fall meeting 2025. New Orleans, USA.
- Vergara, H. & **Robledo, V.** (2025). Investigating the Utility of Satellite Earth Observations to Characterize Mesoscale Convective Systems for Actionable Flash Flood Forecasts. In AGU Fall meeting 2025. Washington DC, USA.
- **Robledo, V.,** Vergara, H. J., Henao, J. J., & Mejia, J. (2024). Multiscale analysis of the environmental precursors of mesoscale convective systems in the Tropical Andes. In AGU Fall Meeting 2024. Washington DC, USA.
- Nikolopoulos, E., Anagnostopoulos, G., Gourley, J. J., ..., **Robledo V.** (2024). Advancing Flash Flood Forecasting Capabilities in West Africa with Machine Learning and Satellite Observations. In 104th AMS Annual Meeting. Baltimore, USA.
- Vergara, H. J., **Robledo, V.,** Anagnostopoulos, G., ..., Nikolopoulos, E. I., (2023). Improving Flash Flood Monitoring and Forecasting Capabilities in West Africa with Satellite Observations and Precipitation Forecasts. In AGU Fall Meeting 2023. San Francisco, USA.
- **Robledo, V.** (2022). "Climatological characteristics and large-scale environments of mesoscale convective systems in northwestern South America using a persistence technique". In AGU Fall Meeting 2022. Chicago, USA.

RESEARCH VISITS

- 2025

NASA Marshall Space Flight Center, Huntsville, Alabama,

Capacity-building workshop on the EF5 model (Presenter).

Host: SERVIR team.

- 2025

AGRHYMET Regional Centre (ARC), Accra, Ghana.

Capacity-building workshop on the EF5 model (Presenter).

Project: NASA's SERVIR Program. Machine learning-based flash flood forecasting in West Africa with satellite observations.

- 2022

Desert Research Institute (DRI), Department of Atmospheric Science, Reno, NV, United States.

Master's short-term internship focused on atmospheric sciences.

Host: Dr. John Mejía. Associate Research Professor.

LANGUAGES

- Spanish (Mothertongue)
- English (Fluent)